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Pedal device

Abstract

A pedal pad in an organ-type pedal device swings about a predetermined axis when stepped on by a driver's foot. A sensor unit outputs a signal corresponding to a swing angle of the pedal pad. A reaction force generation mechanism generates a reaction force against a driver's stepping force applied to the pedal pad. A first housing holds or covers at least one of a rotation shaft provided at the axis of the swinging of the pedal pad, the sensor unit, and the reaction force generation mechanism. A second housing is made of a rigid portion having a Young's modulus greater than that of the first housing or has the rigid portion at least in part. The second housing is configured to support a portion of the reaction force generation mechanism on one side opposite to the pedal pad by the rigid portion.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present application is a continuation application of International Patent Application No. PCT/JP2022/013789 filed on Mar. 24, 2022, which designated the U.S. and claims the benefit of priority from Japanese Patent Application No. 2021-082790 filed on May 14, 2021. The entire disclosures of all of the above applications are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to a pedal device to be mounted on a vehicle.

BACKGROUND

(2) Conventionally, there has been known an organ-type pedal device in which a portion of a pedal pad that is stepped on by a driver, and a coil spring is provided inside a housing as a reaction force generation mechanism for generating a reaction force against a driver's pedaling force applied to the pedal pad.

SUMMARY

(3) According to an aspect of the present disclosure, an organ-type pedal device configured to transmit an electric signal corresponding to a pedal operation amount by a driver to an electronic control device of a vehicle includes a pedal pad, a sensor unit, a reaction force generation mechanism, a first housing and a second housing. The pedal pad is configured to swing about a predetermined axis when being stepped on by a driver's foot. A step portion of the pedal pad to be stepped on by the driver is arranged above the axis in a vertical direction when mounted on the vehicle. The sensor unit is configured to output a signal corresponding to a swing angle of the pedal pad. The reaction force generation mechanism has an elastic member configured to generate a reaction force against a driver's stepping force applied to the pedal pad. The first housing is configured to hold or cover at least one of a rotation shaft provided at the axis of the swinging of the pedal pad, the sensor unit, and the reaction force generation mechanism. The second housing is made of a rigid portion having a Young's modulus greater than that of the first housing or configured to have the rigid portion at least in part. In addition, the second housing is configured to support a portion of the reaction force generation mechanism on one side opposite to the pedal pad by the rigid portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Objects, features, and advantages of the present disclosure will become more apparent from the following detailed description made with reference to the accompanying drawings, in which:
- (2) FIG. 1 is a perspective view of a pedal device according to a first embodiment;
- (3) FIG. 2 is a perspective view of the pedal device according to the first embodiment when being viewed from a direction different from that of FIG. 1;
- (4) FIG. 3 is a perspective view of the pedal device according to the first embodiment, viewed from a second housing side;
- (5) FIG. 4 is a cross-sectional view perpendicular to a pivot axis of a pedal pad in the pedal device according to the first embodiment;
- (6) FIG. 5 is a configuration diagram of a brake-by-wire system using the pedal device according to the first embodiment;
- (7) FIG. 6 is a cross-sectional view perpendicular to a pivot axis of a pedal pad in a pedal device according to a second embodiment;
- (8) FIG. 7 is a cross-sectional view taken along a line VII-VII in FIG. 6;
- (9) FIG. 8 is a diagram for explaining a method of assembling a first housing and a second housing in the pedal device according to the second embodiment;
- (10) FIG. 9 is a diagram for explaining a method of assembling each member in the pedal device according to the second embodiment;
- (11) FIG. 10 is a diagram subsequent to FIG. 9 for explaining a method of assembling each member in the pedal device according to the second embodiment;
- (12) FIG. 11 is a diagram subsequent to FIG. 10 for explaining a method of assembling each member in the pedal device according to the second embodiment;

(13) FIG. 12 is a sectional view showing a portion corresponding to FIG. 7 in a pedal device according to a third embodiment;

(14) FIG. 13A is a diagram for explaining a method of assembling a first housing and a second housing in a pedal device according to the third embodiment;

(15) FIG. 13B is a diagram for explaining a method of assembling a first housing and a second housing in a pedal device according to a modification of the third embodiment;

(16) FIG. 14 is a cross-sectional view perpendicular to a pivot axis of a pedal pad in a pedal device according to a fourth embodiment;

(17) FIG. 15 is a cross-sectional view taken along a line XV-XV of FIG. 14;

(18) FIG. 16 is a sectional view showing a portion corresponding to FIG. 15 in the pedal device according to a fifth embodiment;

(19) FIG. 17 is a cross-sectional view perpendicular to a pivot axis of a pedal pad in a pedal device according to a sixth embodiment;

(20) FIG. 18 is a cross-sectional view taken along line XVIII-XVIII in FIG. 17;

(21) FIG. 19 is a sectional view showing a portion corresponding to FIG. 17 in the pedal device according to a seventh embodiment;

(22) FIG. 20 is a perspective view of a second housing included in a pedal device according to an eighth embodiment;

(23) FIG. 21 is a perspective view of a second housing included in a pedal device according to a ninth embodiment;

(24) FIG. 22 is a cross-sectional view perpendicular to a pivot axis of a pedal pad in a pedal device according to a tenth embodiment;

(25) FIG. 23 is a sectional view perpendicular to a pivot axis of a pedal pad in a pedal device according to an eleventh embodiment;

(26) FIG. 24 is a sectional view perpendicular to a pivot axis of a pedal pad in a pedal device according to a twelfth embodiment;

(27) FIG. 25 is a diagram when being viewed in an XXV direction of FIG. 24;

(28) FIG. 26 is a cross-sectional view showing a method of fixing a first housing and a second housing in a pedal device according to a thirteenth embodiment;

(29) FIG. 27 is a cross-sectional view showing a method of fixing a first housing and a second housing in a pedal device according to a fourteenth embodiment;

(30) FIG. 28 is a sectional view showing a method of fixing a first housing and a second housing in a pedal device according to a fifteenth embodiment;

(31) FIG. 29 is a cross-sectional view showing a method of fixing a first housing and a second housing in a pedal device according to a sixteenth embodiment;

(32) FIG. 30 is a cross-sectional view showing a method of fixing a first housing and a second housing in a pedal device according to a seventeenth embodiment;

(33) FIG. 31 is a sectional view showing a method of fixing a first housing and a second housing in a pedal device according to an eighteenth embodiment;

(34) FIG. 32 is a sectional view showing a method of fixing a first housing and a second housing in the pedal device according to a nineteenth embodiment;

(35) FIG. 33 is a cross-sectional view showing a fixing method and a positioning structure between a first housing and a second housing in the pedal device according to a twentieth embodiment;

(36) FIG. 34 is a diagram showing an example of a positioning structure between the first housing and the second housing in the pedal device according to the twentieth embodiment;

(37) FIG. 35 is a view showing another example of a positioning structure between the first housing and the second housing in the pedal device according to the twentieth embodiment;

(38) FIG. 36 is a view showing another example of a positioning structure between the first housing and the second housing in the pedal device according to the twentieth embodiment;

(39) FIG. 37 is a cross-sectional view showing a fixing method and a positioning structure between

a first housing and a second housing in a pedal device according to a twenty-first embodiment;
(40) FIG. **38** is a perspective view showing a positioning structure between a first housing and a second housing in a pedal device according to a twenty-second embodiment;
(41) FIG. **39** is a cross-sectional view taken along a line XXXIX-XXXIX of FIG. **38**; and
(42) FIG. **40** is a perspective view of a pedal device according to a twenty-third embodiment.

DETAILED DESCRIPTION

(43) In an organ-type pedal device, a portion of a pedal pad that is stepped on by a driver is arranged above a swing axis of the pedal device in the vertical direction when the pedal device is mounted on a vehicle. The organ-type pedal device is used as an accelerator pedal device, a brake pedal device, or the like. In the organ-type accelerator pedal device, a coil spring may be provided inside a housing as a reaction force generation mechanism for generating a reaction force against a driver's pedaling force applied to the pedal pad. A portion of the coil spring opposite to the pedal pad may be supported by an inner wall of the housing.

(44) In the pedal device, when a driver's pedaling force is applied to the pedal pad, the coil spring may bend or deflect. The elastic force of the coil spring acts on the inner wall of the housing as well as the pedal pad. Therefore, if the housing is deformed by the load acting on the inner wall of the housing from the coil spring, there arises a problem that it becomes difficult to generate a stable reaction force from the coil spring to the pedal pad.

(45) By the way, in general, a coil spring used in a brake pedal device has a greater elastic force than a coil spring used in an accelerator pedal device. When the configuration of the pedal device described is applied to the brake pedal device, there may be a greater concern that the above-described problem will occur.

(46) It is an object of the present disclosure to provide a pedal device including a reaction force generation mechanism, which has stabilized pedaling force characteristics by preventing deformation of a housing while restricting an increase in weight.

(47) According to one aspect of the present disclosure, an organ-type pedal device includes a first housing and a second housing, and a portion of a reaction force generation mechanism opposite to a pedal pad is supported by a rigid portion of the second housing. With such configuration, even when a pedaling force is applied to the pedal pad by a driver and a high load is input from the reaction force generation mechanism to the rigid portion, an amount of deformation of the rigid portion is small. Therefore, the reaction force generation mechanism can generate a stable reaction force against the pedal pad. Thus, in such pedal device, by adding a required-minimum member such as a rigid portion to a portion of the pedal device that supports the reaction force generation mechanism, pedaling force characteristics are stabilized and an increase in weight of the pedal device is minimized, thereby making it possible to achieve weight reduction.

(48) Embodiments of the present disclosure will now be described with reference to the drawings. Parts that are identical or equivalent to each other in the following embodiments are assigned the same reference numerals and will not be repeatedly described.

First Embodiment

(49) The first embodiment is described with reference to FIGS. **1** to **5**. A pedal device **1** of the present embodiment is an organ-type pedal device that is mounted on a vehicle and that is stepped on by a driver's pedaling force. The organ-type pedal device **1** refers to a structure in which a portion of a pedal pad **40** that is stepped on by the driver is arranged above a swing axis CL in the vertical direction when mounted on the vehicle. The three-dimensional coordinates shown in FIGS. **1** to **4** indicate an up-down direction, a front-rear direction, and a left-right direction when the pedal device **1** is mounted on the vehicle.

(50) In the organ-type pedal device **1**, a portion of the pedal pad **40** that is in front of the vehicle than the swing axis CL swings toward a floor **2** or a dash panel in an occupant compartment in response to an increase in the pedaling force of the driver applied to the pedal pad **40**. Such a pedal device **1** is used as an accelerator pedal device, a brake pedal device, or the like. In the present

embodiment, a brake pedal device is described as an example of the pedal device **1**. In the following description, the floor **2** or the dash panel of the vehicle to which the pedal device **1** is attached is referred to as a vehicle body.

(51) First, a brake-by-wire system **100** using the pedal device **1** of the present embodiment is described. As shown in FIG. **5**, the brake-by-wire system **100** is a system in which an electronic control unit (hereinafter referred to as an ECU **110**) mounted on the vehicle drives a brake circuit **120** based on an electric signal output from a sensor unit **50** of the pedal device **1**. In such brake-by-wire system **100**, the brake circuit **120** generates hydraulic pressure required for braking the vehicle to drive wheel cylinders **131** to **134**.

(52) In the brake-by-wire system **100** illustrated in FIG. **5**, the ECU **110** is made of a first ECU **111** and a second ECU **112**. Also, the brake circuit **120** includes a first brake circuit **121** and a second brake circuit **122**.

(53) An electric signal output from the sensor unit **50** of the pedal device **1** is transmitted to the first ECU **111** and the second ECU **112**. The first ECU **111** has a computer, a drive circuit, and the like (not shown). The first ECU **111** supplies electric power to a motor **123** of the first brake circuit **121** and the like to drive and control the first brake circuit **121**. The second ECU **112** also has a computer and a drive circuit, and the like (not shown). The second ECU **112** drives and controls an electromagnetic valve, a motor, and the like (not shown) of the second brake circuit **122**.

(54) The first brake circuit **121** has a reservoir **124**, a motor **123**, a gear mechanism **125**, a master cylinder **126** and the like. The reservoir **124** stores brake fluid. The motor **123** drives the gear mechanism **125**. The gear mechanism **125** reciprocates a master piston **127** of the master cylinder **126** in an axial direction of the master cylinder **126**. Due to movement of the master piston **127**, the hydraulic pressure of the brake fluid supplied from the reservoir **124** to the master cylinder **126** is increased, and the hydraulic pressure is supplied from the first brake circuit **121** to the second brake circuit **122**.

(55) The second brake circuit **122** is a circuit for executing a normal control, an ABS control, a VSC control, etc. by controlling the hydraulic pressure supplied to each of wheel cylinders **131** to **134** according to the control signal from the second ECU **112**. ABS stands for Anti-lock Braking System, and VSC stands for Vehicle Stability Control. The wheel cylinders **131** to **134** arranged on each wheel drive brake pads provided on each wheel.

(56) When a driver in a vehicle performs a step-on operation to the pedal pad **40** of the pedal device **1**, a signal corresponding to a swing angle (that is, pedal operation amount) of the pedal pad **40** is output from the sensor unit **50** to the first ECU **111** and the second ECU **112**. The first ECU **111** drives the motor **123** to decelerate the vehicle. As a result, when a rotation speed of the motor **123** increases, the master cylinder **126** increases a hydraulic pressure of the brake fluid supplied from the reservoir **124**. The resultant hydraulic pressure of the brake fluid is transmitted from the first brake circuit **121** to the second brake circuit **122**.

(57) The second ECU **112** performs the normal control, the ABS control, the VSC control, and the like. For example, the second ECU **112** controls driving of each solenoid valve of the second brake circuit **122** in the normal control for braking according to the operation of the pedal pad **40** by the driver. The second ECU **112** then causes the hydraulic pressure supplied from the first brake circuit **121** to be supplied to each of the wheel cylinders **131** to **134** via the second brake circuit **122**. Therefore, the brake pads driven by the respective wheel cylinders **131** to **134** each come into frictional contact with the corresponding brake disc, thereby a brake is applied to each wheel, and the vehicle decelerates.

(58) Also, for example, the second ECU **112** calculates a slip ratio of each wheel based on the speed of each wheel of the vehicle and the vehicle speed, and performs ABS control based on the calculation result of the slip rate. Under the ABS control, the hydraulic pressure supplied to each of the wheel cylinders **131** to **134** is adjusted, to prevent each wheel from being locked. Also, for example, the second ECU **112** calculates a sideslip state of the vehicle based on a yaw rate, a

steering angle, an acceleration, a speed of each wheel, a vehicle speed, etc., and performs VSC control based on the calculation result. Under the VSC control, a control target wheel for stabilizing the vehicle turning is selected, and the hydraulic pressure of the wheel cylinders **131** to **134** corresponding to the selected wheel(s) is raised, to prevent the sideslip of the vehicle. Thus, the vehicle can travel stably. Further to the normal control, ABS control, and VSC control described above, the second ECU **112** may perform a collision avoidance control, a regenerative cooperation control, and the like based on signals from other ECUs (not shown).

(59) Next, the pedal device **1** is explained. As shown in FIGS. **1** to **4**, the pedal device **1** includes a first housing **10**, a second housing **20**, a reaction force generation mechanism **30**, the pedal pad **40**, the sensor unit **50**, and the like.

(60) The first housing **10** is a member that holds or covers at least one of a rotation shaft **41** of the pedal pad **40**, the sensor unit **50** and the reaction force generation mechanism **30**. The first housing **10** of the present embodiment is made of resin, for example. The first housing **10** is shaped like a box having a top wall **11**, a left-side wall **12**, a right-side wall **13**, a front wall **14**, and the like. A space for arranging the sensor unit **50**, the reaction force generation mechanism **30**, and the like is provided inside the first housing **10** of the present embodiment.

(61) The first housing **10** has an opening **15** into which the reaction force generation mechanism **30** can be inserted in at least one direction. Specifically, the first housing **10** of the present embodiment has the largest opening **15** on one side facing the vehicle body on which the pedal device **1** is installed. It can also be said that the first housing **10** of the present embodiment has the largest opening **15** on one side of the pedal pad **40** in a swing direction. The largest opening **15** of the first housing **10** has a size that allows an elastic member of the reaction force generation mechanism **30** to be inserted without bending during a manufacturing process of the pedal device **1**. Further, the largest opening **15** of the first housing **10** is configured to be closed by the second housing **20**. Thus, the first housing **10** of the present embodiment covers the reaction force generation mechanism **30** together with the second housing **20**.

(62) Further, the first housing **10** of the present embodiment has a cylindrical bearing portion **16** that rotatably supports the rotation shaft **41** that is provided on the swing axis CL of the pedal pad **40**. The rotation shaft **41** of the pedal pad **40** is a columnar shaft rotatably provided in the bearing portion **16**. The sensor unit **50** for detecting a rotation angle of the rotation shaft **41** is provided on or about the axis CL of the rotation shaft **41** of the pedal pad **40**. Therefore, the first housing **10** of the present embodiment holds and covers the rotation shaft **41** of the pedal pad **40** and a part of the sensor unit **50**.

(63) Since the sensor unit **50** is provided on or about the axis CL of the rotation shaft **41**, it is possible to directly detect the rotation angle of the rotation shaft **41**. As the sensor unit **50**, for example, it is possible to adopt a non-contact sensor circuit using a Hall IC or a magneto resistive element, or a contact sensor circuit. The sensor unit **50** outputs a signal corresponding to the rotation angle of the rotation shaft **41** from a connector **51** provided outside the first housing **10** to the ECU **110** of the vehicle. The rotation angle of the rotation shaft **41** and the swing angle of the pedal pad **40** (that is, the pedal operation amount) are the same.

(64) The second housing **20** is provided on one side of the first housing **10** in a direction in which the pedal pad **40** swings. The second housing **20** is provided to close the largest opening **15** of the first housing **10**. The second housing **20** of the present embodiment extends continuously from a portion of the first housing **10** on the front side of the vehicle to a portion on the rear side of the vehicle. The first housing **10** and the second housing **20** are fixed by, for example, a fixing member such as a bolt **60** inserted from a surface of the second housing **20** on a vehicle body side to a first housing **10** side. The method of fixing the first housing **10** and the second housing **20** is not limited to the above, and various methods can also be adopted as described later in the thirteenth to twenty-second embodiments.

(65) The second housing **20** is made of a member having a Young's modulus greater than that of the

first housing **10**, or has a member having a Young's modulus greater than that of the first housing **10** at least in part. In the following description, a member having a Young's modulus greater than that of the first housing **10** is referred to as a "rigid portion **21**." The rigid portion **21** can be made of metal, for example. The rigid portion can also be rephrased as a high-strength portion. The second housing **20** of the present embodiment is entirely made of the rigid portion **21**. The second housing **20** of the present embodiment can be manufactured, for example, by pressing a metal material, sheet metal working, cold forging, warm forging, or the like.

(66) A configuration in which the second housing **20** has the rigid portion **21** as a part of a resin portion is described later in the eleventh and twelfth embodiments.

(67) The second housing **20** is arranged at a position between (i) an opposite portion of the reaction force generation mechanism **30** opposite to the pedal pad **40** and (ii) the vehicle body. The second housing **20** supports a portion of the reaction force generation mechanism **30** opposite to the pedal pad **40**. In the present embodiment, the portion of the reaction force generation mechanism **30** opposite to the pedal pad **40** corresponds to one end **311** of a leaf spring **31** of the reaction force generation mechanism **30**. Since the second housing **20** of the present embodiment is entirely made of the rigid portion **21**, even when a high load is input from the reaction force generation mechanism **30**, an amount of deformation of a portion **22** of the second housing **20** supporting the reaction force generation mechanism **30** is small. Note that the portion **22** that supports the reaction force generation mechanism **30** may also be called as a spring receiving portion **22**.

(68) Further, the second housing **20** also integrally has a portion **23** that is fixed to the vehicle body, in addition to the portion **22** that supports the reaction force generation mechanism **30**. In the second housing **20**, the portion **22** supporting the reaction force generation mechanism **30** and the portion **23** that is fixed to the vehicle body are integrally formed by the rigid portion **21**. The portion **23** fixed to the vehicle body extends leftward and rightward from the first housing **10**, and is provided with a hole **24** piercing in a plate thickness direction. The second housing **20** is fixed to the floor **2** or the dash panel of the vehicle by inserting a bolt (not shown) through the hole **24**. The dash panel is a partition wall that separates the interior of the vehicle from the exterior such as the engine room of the vehicle, and is sometimes called as a bulkhead.

(69) The pedal pad **40** is made of, for example, metal or resin in a plate-like shape, and is arranged obliquely with respect to the floor **2**. Specifically, the pedal pad **40** is obliquely arranged so that an upper end thereof faces the front of the vehicle and a lower end thereof faces the rear of the vehicle. A thick portion **42** is provided on an upper portion of the pedal pad **40** as a portion to be stepped on by the driver. The thick portion **42** is arranged above the swing axis CL of the pedal pad **40** in the vertical direction when the pedal pad **40** is mounted on the vehicle. The arrangement of the pedal pad **40** is not limited to the one shown in FIGS. **1** to **4**, that is, may also be arranged substantially perpendicular to the floor **2**, for example.

(70) A connecting plate **43** is provided on a back surface of the pedal pad **40**. The connecting plate **43** integrally has a back plate portion **44** fixed to a back surface of the pedal pad **40** and a side plate portion **45** substantially perpendicular to the back plate portion **44**. The side plate portion **45** of the connecting plate **43** is fixed to the rotation shaft **41**. As described above, the rotation shaft **41** is rotatably supported by the bearing portion **16** of the first housing **10**. Therefore, the pedal pad **40** swings about the axis CL of the rotation shaft **41** in a forward and reverse rotation directions within a predetermined angle range when the pedal pad **40** is stepped on by the driver's foot. By configuring the connecting plate **43** to have the back plate portion **44** and the side plate portion **45**, the pedal pad **40** and the rotation shaft **41** are arranged in separated positions, and the sensor unit **50** is easily arrangeable in a space close to the rotation shaft **41**.

(71) The reaction force generation mechanism **30** is a mechanism that generates a reaction force against the driver's pedaling force applied to the pedal pad **40**. Since the pedal device **1** includes the reaction force generation mechanism **30**, even if a mechanical connection between the pedal pad **40** and the master cylinder **126** is eliminated, it is possible to obtain a reaction force similar to that of a

conventional brake system. Note that the conventional brake system refers to a configuration in which the pedal pad and the master cylinder are mechanically connected, and the pedal pad receives a hydraulic reaction force from the master cylinder.

(72) In the present embodiment, the reaction force generation mechanism **30** has a plurality of elastic members. Specifically, the reaction force generation mechanism **30** has a leaf spring **31**, a large-diameter coil spring **33**, and a small-diameter coil spring **34** as a plurality of elastic members. By configuring the reaction force generation mechanism **30** to have a plurality of elastic members, the reaction force of the reaction force generation mechanism **30** is changeable in multiple stages with respect to changes in the swing angle of the pedal pad **40** (that is, the amount of pedal operation). As a result, the reaction force generation mechanism **30** can reproduce the multistage reaction force characteristics that are unique to the conventional brake system.

(73) The leaf spring **31** is bent to form a convex curved surface toward the floor **2** when not receiving a load. The one end **311** of the leaf spring **31** (that is, a portion of the reaction force generation mechanism **30** opposite to the pedal pad **40**) is arranged at a position between the rotation shaft **41** of the pedal pad **40** and the second housing **20**. The one end **311** of the leaf spring **31** is fixed to the portion **22** of the second housing **20** that supports the reaction force generation mechanism **30**.

(74) On the other hand, at an other end **312** of the leaf spring **31**, a lower holder **35**, the large-diameter coil spring **33**, a spring seat **36**, a small-diameter coil spring **34**, an upper holder **37**, a connecting rod **70**, etc. are provided in this order from a leaf spring **31** side. The lower holder **35** is fixed to the other end **312** of the leaf spring **31**. The large-diameter coil spring **33** has one end on a leaf spring **31** side supported by the lower holder **35** and other end on a pedal pad **40** side supported by the spring seat **36**. The small-diameter coil spring **34** has one end on a leaf spring **31** side supported by the spring seat **36** and other end on a pedal pad **40** side supported by the upper holder **37**. The connecting rod **70** has one end on a pedal pad **40** side fixed to the pedal pad **40**, and other end on a leaf spring **31** side slidably contacts the upper holder **37**. The connecting rod **70** may also be configured such that one end on a pedal pad **40** side is swingably connected to the pedal pad **40** and other end on a leaf spring **31** side swingably connected to the upper holder **37**. The connecting rod **70** is inserted through an upper opening **17** provided in the first housing **10**. The upper opening **17** is sized to allow the small-diameter coil spring **34** and the upper holder **37** to pass therethrough.

(75) With such configuration, when the driver applies a pedaling force to the pedal pad **40** and the pedal pad **40** swings toward the first housing **10** and the second housing **20**, a load is applied to various members of the reaction force generation mechanism **30** from the pedal pad **40** via the connecting rod **70**. Therefore, the leaf spring **31**, the large-diameter coil spring **33**, and the small-diameter coil spring **34**, which constitute the reaction force generation mechanism **30**, bend according to their respective spring constants, and generate a reaction force against the pedaling force applied to the pedal pad **40** by the driver. Specifically, when a load is applied to each of the members of the reaction force generation mechanism **30** from the pedal pad **40** via the connecting rod **70**, the large-diameter coil spring **33** and the small-diameter coil spring **34** bend in their spring axial directions. Further, the leaf spring **31** bends so that a portion thereof on an other end **312** side to which the lower holder **35** is fixed comes closer to the second housing **20**. The configurations of the reaction force generation mechanism **30** and the connecting rod **70** are not limited to those illustrated above, and various configurations are adoptable.

(76) A covering member **71** is provided around the connecting rod **70**. The covering member **71** is made of rubber, for example, and is formed in a tubular and bellows shape. On one side of a cylindrical shape, the covering member **71** engages with a groove **72** provided in the middle of the connecting rod **70**, and on other side of the cylindrical shape, the covering member **71** engages with the upper opening **17** of the first housing **10**. The covering member **71** prevents foreign substances, water, and the like from entering the inside of the first housing **10** through the upper opening **17** of the first housing **10**.

(77) The pedal device **1** shown in FIGS. **1** to **4** is, respectively in a state where the pedaling force of the driver is not applied to the pedal pad **40** (that is, the initial position of the pedal pad **40**). An initial position of the pedal pad **40** is regulated by a full-close stopper (not shown).

(78) Although not shown, in the pedal device **1**, the pedal pad **40** swings about the axis CL of the rotation shaft **41** when a driver's pedaling force is applied to the pedal pad **40**. Then, the portion of the pedal pad **40** that is above in the vehicle with respect to the axis CL moves toward the floor **2** or the dash panel. At this time, the sensor unit **50** outputs a signal corresponding to the rotation angle of the rotation shaft **41** (that is, the swing angle of the pedal pad **40**) to the ECU **110** of the vehicle. The ECU **110** drives and controls the brake circuit **120** to generate hydraulic pressure required for braking the vehicle, and the hydraulic pressure drives the brake pads to decelerate or stop the vehicle.

(79) Next, a method for manufacturing the pedal device **1** of the first embodiment is described. In the method of manufacturing the pedal device **1** of the first embodiment, first, the rotation shaft **41** and the pedal pad **40** are assembled to the first housing **10** to form a first sub-assembly. On the other hand, the reaction force generation mechanism **30** is assembled to the second housing **20** to form a second sub-assembly. Then, the first housing **10** and the second housing **20** are fixed with a fixing member such as a bolt **60**, and the first sub-assembly and the second sub-assembly are assembled. At this time, the plurality of elastic members of the reaction force generation mechanism **30** assembled to the second housing **20** are arrangeable to the inside of the first housing **10** from the largest opening **15** of the first housing **10** without bending. Thereafter, the first housing **10** and the second housing **20** may be fixed. To bend the plurality of elastic members arranged inside the first housing **10**, for example, the elastic members may be pressed by the connecting rod **70** attached to the pedal pad **40**, or the connecting rod **70** attached to the upper holder **37** may be pressed by the pedal pad **40**. Alternatively, a jig or the like may be used to bend the plurality of elastic members. Thus, assembly of the pedal device **1** of the first embodiment is easily performable to have each of the various members assembled in position, which is achieved by the improved assemblability of each of those members in the manufacturing process.

(80) The pedal device **1** of the first embodiment described above has the following effects.

(81) (1) In the first embodiment, the pedal device **1** has a configuration in which the rigid portion **21** of the second housing **20** having a Young's modulus greater than that of the first housing **10** supports the reaction force generation mechanism **30** on the opposite side relative to the pedal pad **40**. As a result, even when the driver applies a pedaling force to the pedal pad **40** and a high load is input from the reaction force generation mechanism **30** to the second housing **20**, the amount of deformation of the second housing **20** formed by the rigid portion **21** is small. Therefore, the reaction force generation mechanism **30** can generate a stable reaction force toward the pedal pad **40**. Therefore, in the pedal device **1**, by adding the required-minimum members such as the second housing **20** to the portion **22** that supports the reaction force generation mechanism **30**, the pedaling force characteristics are stabilized and the weight increase of the pedal device **1** is minimized, thereby reducing the weight thereof.

(82) (2) In the first embodiment, the second housing **20** made up of the rigid portion **21** is arranged at a position between (i) the portion of the reaction force generation mechanism **30** opposite to the pedal pad **40** (the one end **311** of the leaf spring **31** in the first embodiment) and (ii) the vehicle body. According to the above, even when a pedaling force is applied to the pedal pad **40** by the driver and a high load is input to the second housing **20** from the reaction force generation mechanism **30**, the amount of deformation of the second housing **20** constituted by the rigid portion **21** is small.

(83) In the first embodiment, the second housing **20** integrally has the portion **23** that is fixed to the vehicle body, in addition to the portion **22** that supports the reaction force generation mechanism **30**. According to the above, by fixing the pedal device **1** and the vehicle body using the portion **23** of the second housing **20** that is fixed to the vehicle body, deformation of the pedal device **1** when

mounted on the vehicle is suppressible. Therefore, it is possible to stabilize the pedaling force characteristics and to minimize an increase in the weight of the pedal device **1**, thereby achieving weight reduction.

(84) (3) In the first embodiment, the second housing **20** is made of metal. According to the above, metal is exemplified as the rigid portion **21** having a Young's modulus greater than that of the first housing **10**.

(85) (4) In the first embodiment, the first housing **10** has the opening **15** into which the reaction force generation mechanism **30** can be inserted in at least one direction, and covers the reaction force generation mechanism **30** together with the second housing **20** covering the opening **15**. According to the above, it is possible to prevent foreign matter from entering the reaction force generation mechanism **30** arranged inside the first housing **10**.

(86) (5) In the first embodiment, the first housing **10** has the bearing portion **16** that rotatably supports the rotation shaft **41** of the pedal pad **40**. According to the above, even when the pedaling force is applied to the pedal pad **40** by the driver and a high load is input from the reaction force generation mechanism **30** to the second housing **20**, transmission of the load from the second housing **20** to the first housing **10** is divided. Therefore, the first housing **10** is less susceptible to the high load, which may be input from the reaction force generation mechanism **30** to the second housing **20**. Therefore, by providing the bearing portion **16** in the first housing **10**, the swinging movement of the pedal pad **40** is stabilized, and the reliability of the output signal of the sensor unit **50** is improvable. Further, in the manufacturing process of the pedal device **1**, the rotation shaft **41** and the pedal pad **40** are assembled to the first housing **10**, to form the first sub-assembly, while the second housing **20** and the reaction force generation mechanism **30** are assembled to form the second sub-assembly. Then, by assembling the first sub-assembly and the second sub-assembly, it becomes possible to easily assemble the pedal device **1**, and the ease of assembly of the pedal device **1** is improvable.

(87) (6) In the first embodiment, the first housing **10** has the largest opening **15** in which the elastic member of the reaction force generation mechanism **30** can be inserted without bending on the side where the second housing **20** is provided, and the largest opening **15** thereof is closed by the second housing **20**. According to the above, in the manufacturing process of the pedal device **1**, the elastic member of the reaction force generation mechanism **30** without being bent is arranged at a position between the first housing **10** and the second housing **20** from the largest opening **15** of the first housing **10**. Therefore, when assembling the reaction force generation mechanism **30** to a position between the first housing **10** and the second housing **20**, a jig for bending the elastic member of the reaction force generation mechanism **30** is not required. Therefore, the pedal device **1** having various members can be easily assembled, and the assemblability is improvable.

(88) (7) In the first embodiment, the reaction force generation mechanism **30** has a plurality of elastic members. According to the above, it is possible to change the reaction force of the reaction force generation mechanism **30** in multiple stages with respect to the change in the pedal operation amount, so that the multistage reaction force characteristic specific to the conventional brake system can be reproducible.

(89) (8) In the first embodiment, the multiple elastic members include at least one leaf spring **31**. The leaf spring **31** is fixed to the rigid portion **21** of the second housing **20**. If, in case that the elastic member of the reaction force generation mechanism **30** is composed only of a coil spring, a space corresponding to "diameter of coil spring×number of turns" and "deflection amount" is required. On the other hand, when the leaf spring **31** is used as the elastic member of the reaction force generation mechanism **30**, it is sufficient to have a space corresponding to the "thickness of the leaf spring **31**" and the "deflection amount." Therefore, the space required for installing the leaf spring **31** is smaller than the space required for installing the coil spring. Thus, by using the leaf spring **31** as the elastic member of the reaction force generation mechanism **30**, the pedal device **1** can reduce the volume inside the first housing **10**. As a result, the pedal device **1** is downsized.

(90) (9) The pedal device **1** of the first embodiment is a brake pedal device used in the brake-by-wire system **100** in which a brake circuit generates hydraulic pressure required for braking the vehicle by drive control of the ECU **110** based on the output signal of the sensor unit **50**. According to the above, by using the pedal device **1** in the brake-by-wire system **100**, accurate vehicle braking control is realized by the drive control of the ECU **110** based on the electric signal with high detection accuracy which is output from the sensor unit **50** of the pedal device **1**.

Second Embodiment

(91) The second embodiment is described with reference to FIGS. **6** to **11**. In the second embodiment, the configuration of a reaction force generation mechanism **30** is changed with respect to the first embodiment, while the rest of the configuration stay is unchanged therefrom, thereby focusing the description only on such change from the first embodiment.

(92) As shown in FIGS. **6** and **7**, a pedal device **1** of the second embodiment also includes a first housing **10**, a second housing **20**, a reaction force generation mechanism **30**, a pedal pad **40**, a sensor unit, and the like. In each of the drawings referred to in or after the second embodiment, the configuration is shown in a simplified manner, and illustration of the sensor unit is omitted.

(93) The first housing **10** has a largest opening **15** provided on one side facing the vehicle body, on which the pedal device **1** is installed. The opening **15** of the first housing **10** is closed by the second housing **20**. The second housing **20** is made of a rigid portion **21** having a Young's modulus greater than that of the first housing **10**. The second housing **20** of the second embodiment is also composed entirely of the rigid portion **21**.

(94) The reaction force generation mechanism **30** has one elastic member and an upper holder **37**. The one elastic member is made of a coil spring **32**. A lower end portion of the coil spring **32** opposite to the pedal pad **40** is supported by the second housing **20**. In the second embodiment, a lower end portion **321** of the coil spring **32** corresponds to a portion of the reaction force generation mechanism **30** opposite to the pedal pad **40**. That is, the second housing **20** supports a portion of the reaction force generation mechanism **30** opposite to the pedal pad **40**. Also in the second embodiment, the second housing **20** is entirely made of the rigid portion **21**. Therefore, even when a high load is input from the reaction force generation mechanism **30**, the amount of deformation of the portion **22** of the second housing **20** that supports the reaction force generation mechanism **30** is small.

(95) The upper holder **37** is provided at an upper end portion **322** of the coil spring **32** on the pedal pad **40** side. The upper holder **37** and the pedal pad **40** are connected by a connecting rod **70**. The connecting rod **70** has one end on a pedal pad **40** side fixed to the pedal pad **40** and another end on a leaf spring **31** side slidably contacting the upper holder **37**. The connecting rod **70** may also be configured such that one end on a pedal pad **40** side is swingably connected to the pedal pad **40** and other end on a leaf spring **31** side swingably connected to the upper holder **37**. The connecting rod **70** is inserted through the upper opening **17** provided in the first housing **10**. In the second embodiment, the upper opening **17** is formed to be smaller than an outer diameter of the upper holder **37**. In other words, the upper opening **17** has a size through which the connecting rod **70** is insertable, but a size through which the upper holder **37** is not insertable.

(96) Next, a method for manufacturing the pedal device **1** of the second embodiment is described. As shown in FIG. **8**, in the pedal device **1** of the second embodiment, the rotation shaft **41** and the pedal pad **40** are assembled to the first housing **10** to form the first sub-assembly, like the first embodiment. On the other hand, the reaction force generation mechanism **30** is assembled to the second housing **20** to form the second sub-assembly. Then, as indicated by an arrow A in FIG. **8**, the reaction force generation mechanism **30** is inserted through the opening **15** of the first housing **10**, and the first sub-assembly and the second sub-assembly are assembled. It should be noted that the pedal pad **40** is omitted in FIG. **8**.

(97) Now, the details of the method of manufacturing the pedal device **1** of the second embodiment are described with reference to FIGS. **9** to **11**. First, as shown in FIG. **9**, the coil spring **32** of the

reaction force generation mechanism **30** is installed on the second housing **20**, and the upper holder **37** is installed on the coil spring **32**. The second sub-assembly is composed in such manner. At this time, the coil spring **32** is substantially in a free length state. It should be noted that there is no problem even if the coil spring **32** is slightly bent.

(98) Next, as shown in FIG. **10**, the first sub-assembly in which the rotation shaft **41** and the pedal pad **40** are assembled with the first housing **10** is brought to the above of the second sub-assembly. Then, as shown by arrows B and C in FIG. **10**, while the coil spring **32** and the upper holder **37** are inserted inside the first housing **10** and the coil spring **32** is bent, the housing **10** is moved toward the second housing **20** until it touches the second housing **20**. At this time, the second housing **20** may be moved toward the first housing **10**.

(99) Subsequently, as shown in FIG. **11**, the first housing **10** and the second housing **20** are fixed with fixing members such as bolts, and the first sub-assembly and the second sub-assembly are assembled. Thus, each of the members of the pedal device **1** of the second embodiment is also easily assembled, thereby the assemblability of each of the members in the manufacturing process is improvable.

(100) The pedal device **1** of the second embodiment described above achieves the same effects as the first embodiment due to the corresponding configuration as the first embodiment. Further, in the second embodiment, the bending direction of the elastic member of the reaction force generation mechanism **30** and the assembling direction of the first housing **10** and the second housing **20** are both in parallel with a virtual plane perpendicular to the axis CL. Therefore, in the manufacturing process of the pedal device **1**, assembling the first housing **10** and the second housing **20** is complete, while the elastic member of the reaction force generation mechanism **30** is bent.

(101) As a modification of the second embodiment, although not shown in any drawings, a method of bending the coil spring **32** when moving the first housing **10** toward the second housing **20** may also include the following, other than the method shown in FIG. **10**. For example, in one case, the pedal pad **40** may be attached to the connecting rod **70** in advance, and the lower end of the connecting rod **70** may be used to press the upper holder **37**, for bending the coil spring **32**.

Alternatively, a method of bending the coil spring **32** may be, for example, attaching the connecting rod **70** to the upper holder **37** in advance, and the pedal pad **40** may be used to press an upper end portion of the connecting rod **70**, for bending the coil spring **32**.

Third Embodiment and its Modification

(102) The third embodiment and its modification are described with reference to FIGS. **12**, **13A**, and **13B**. The third embodiment and its modification have a change in the configuration of the first housing **10** and the second housing **20** with respect to the first embodiment and the like, and the rest is the same as that of first embodiment and the like. Therefore, only parts different from the first embodiment and the like is described. FIGS. **12** and **13A** show the third embodiment, and FIG. **13B** shows a modification of the third embodiment. As is described later, the third embodiment and its modification are the same except for the direction in which the second sub-assembly is assembled to the first sub-assembly.

(103) As shown in FIGS. **12**, **13A**, and **13B**, in the third embodiment and its modification, a first housing **10** in a pedal device **1** has an opening **15** formed as a part of a right-side wall **13** and a bottom wall. The bottom wall of the first housing **10** is a wall thereof facing the vehicle body on which the pedal device **1** is installed.

(104) On the other hand, a second housing **20** is formed in an L shape and is configured to close the opening **15** of the first housing **10**. The second housing **20** is made of a rigid portion **21** having a Young's modulus greater than that of the first housing **10**. The second housing **20** of the third embodiment and its modification is also composed entirely of the rigid portion **21**.

(105) A reaction force generation mechanism **30** has one elastic member and an upper holder **37**. One elastic member is made of a coil spring **32**. A lower end portion of the coil spring **32** opposite

to a pedal pad **40** is supported by the second housing **20**. In the third embodiment, a lower end portion **321** of the coil spring **32** corresponds to a portion of the reaction force generation mechanism **30** opposite to the pedal pad **40**. That is, the second housing **20** supports a portion of the reaction force generation mechanism **30** opposite to the pedal pad **40**. Since the second housing **20** is made of a rigid portion, even when a high load is input from the reaction force generation mechanism **30**, the amount of deformation of the second housing **20** is small.

(106) Next, a method of manufacturing the pedal device **1** according to the third embodiment and its modification is described. As shown in FIGS. **13A** and **13B**, in the pedal device **1** of the third embodiment and its modification, a rotation shaft **41** and the pedal pad **40** are assembled to the first housing **10** in the same manner as in the first embodiment, to compose the first sub-assembly. On the other hand, the reaction force generation mechanism **30** is assembled to the second housing **20** to compose the second sub-assembly. Further, as indicated by an arrow **D1** in FIG. **13A** or by an arrow **D2** in FIG. **13B**, the first sub-assembly and the second sub-assembly are assembled while the reaction force generation mechanism is inserted from under or from a side of the opening **15** of the first housing **10**. Note that the pedal pad **40** is omitted in FIGS. **13A** and **13B**.

(107) The pedal device **1** of the third embodiment and its modification described above can also achieve the same effects as the first embodiment and the like, based on the same or similar configuration as the first embodiment and the like.

Fourth to Tenth Embodiments

(108) In the fourth to tenth embodiments described below, the configuration regarding a rotation shaft **41** and a bearing portion **16** of a pedal pad **40** is mainly described with respect to the change from the first embodiment and the like, while the other parts are similar to those of the first embodiment and the like. Therefore, in the fourth to tenth embodiments, only the parts different from the first embodiment are mainly described.

Fourth Embodiment

(109) As shown in FIGS. **14** and **15**, in the fourth embodiment, a bearing portion **16** that supports a rotation shaft **41** of a pedal pad **40** is provided in a first housing **10** just like the first embodiment. Specifically, the bearing portion **16** provided in the first housing **10** supports one end and the other end of the rotation shaft **41** in an axial center CL direction. The pedal pad **40** is fixed to a central portion of the rotation shaft **41**.

(110) Also in the fourth embodiment, the first housing **10** has an opening **15** in a part of the bottom wall. The opening **15** of the first housing **10** is closed by a second housing **20** made of a rigid portion **21**. The second housing **20** has a spring receiving portion **27** for installing a coil spring **32**. The second housing **20** supports a portion of the coil spring **32** of a reaction force generation mechanism **30** opposite to the pedal pad **40**. Therefore, even when a high load is input from the reaction force generation mechanism **30**, the amount of deformation of the second housing **20** is small.

(111) In a pedal device **1** of the fourth embodiment described above, similarly to the first embodiment and the like, when a high load is input from the reaction force generation mechanism **30** to the second housing **20**, transmission of the load from the second housing **20** to the first housing **10** is divided. Therefore, the first housing **10** is less susceptible to the high load, which may be input from the reaction force generation mechanism **30** to the second housing **20**. Therefore, by providing the bearing portion **16** in the first housing **10**, the swinging movement of the pedal pad **40** is stabilized, and the reliability of the output signal of the sensor unit is improvable.

(112) Further, in the manufacturing process of the pedal device **1**, the rotation shaft **41** and the pedal pad **40** are assembled to the first housing **10**, to form the first sub-assembly, while the second housing **20** and the reaction force generation mechanism **30** are assembled to form the second sub-assembly. Then, by assembling the first sub-assembly and the second sub-assembly, it becomes possible to easily assemble the pedal device **1**, and the ease of assembly of the pedal device **1** is

improvable.

(113) Thus, the pedal device **1** of the fourth embodiment can also achieve the same effects as the first embodiment and the like from the corresponding configuration as the first embodiment and the like.

Fifth Embodiment

(114) The fifth embodiment is a modification of the fourth embodiment. As shown in FIG. **16**, in the fifth embodiment as well, a first housing **10** is provided with a bearing portion **16** that supports a rotation shaft **41** of a pedal pad **40**. Specifically, the bearing portion **16** provided in the first housing **10** supports a central portion of the rotation shaft **41**. The pedal pad **40** is fixed to the one end and the other end of the rotation shaft **41** in the axial center CL direction.

(115) The pedal device **1** of the fifth embodiment achieves the same effects as the first embodiment and the like from the corresponding configuration as in the first embodiment and the like.

Sixth Embodiment

(116) As shown in FIGS. **17** and **18**, in the sixth embodiment, a second housing **20** is provided with a bearing portion **26** that supports a rotation shaft **41** of a pedal pad **40**. Specifically, the bearing portion **26** provided in the second housing **20** supports one end and the other end of the rotation shaft **41** in the axial center CL direction. The pedal pad **40** is fixed to a central portion of the rotation shaft **41**.

(117) Further, the second housing **20** extends from the bearing portion **26** toward the front of the vehicle, and has a spring receiving portion **27** for installing the coil spring **32** in the middle thereof. The coil spring **32** is an example of an elastic member of the reaction force generation mechanism **30**. That is, the second housing **20** supports a portion of the elastic member of the reaction force generation mechanism **30** on one side opposite to the pedal pad **40**.

(118) A first housing **10** covers the coil spring **32** as an elastic member of the reaction force generation mechanism **30** and an upper holder **37** provided at an upper end of the coil spring **32**. The first housing **10** has a largest opening **15** on one side facing the vehicle body, on which a pedal device **1** is installed. Further, the largest opening **15** of the first housing **10** is configured to be closed by the second housing **20**. Note that, in the sixth embodiment, the first housing **10** does not cover the bearing portion **26** and the rotation shaft **41**.

(119) The sixth embodiment described above also achieves the same effects as the first embodiment and the like from the corresponding configuration as the first embodiment and the like. Further, in the pedal device **1** of the sixth embodiment, the second housing **20** has the bearing portion **26**, and supports a portion of the reaction force generation mechanism **30** opposite to the pedal pad **40**. According to the above, it is possible to improve the positional accuracy of a distance between the swing axial center CL of the pedal pad **40** and the reaction force generation mechanism **30**. Therefore, variations in the reaction force of the reaction force generation mechanism **30** due to assembly tolerances of the various members of the pedal device **1** are suppressible.

Seventh Embodiment

(120) The seventh embodiment is a modification of the sixth embodiment. As shown in FIG. **19**, in the seventh embodiment, a second housing **20** is provided with a bearing portion **26** that supports a rotation shaft **41** of a pedal pad **40**. Specifically, the bearing portion **26** provided in the second housing **20** supports a central portion of the rotation shaft **41**. The pedal pad **40** is fixed to the one end and the other end of the rotation shaft **41** in the axial center CL direction.

(121) A pedal device **1** of the seventh embodiment can also achieve the same effects as the sixth embodiment and the like from the same corresponding configuration as the sixth embodiment and the like.

Eighth Embodiment

(122) The eighth embodiment is also a modification of the sixth embodiment. FIG. **20** shows only a second housing **20** included in a pedal device **1** of the eighth embodiment. The second housing **20**

has a bearing portion **26** that supports a rotation shaft **41** of a pedal pad **40**. Specifically, the bearing portion **26** provided in the second housing **20** is configured to support the one end and the other end of the rotation shaft **41** in the axial center CL direction.

(123) Further, the second housing **20** has a portion **22** (i.e., a spring receiving portion **22**) that supports a one end **311** of a leaf spring **31** that constitutes a reaction force generation mechanism **30**. The spring receiving portion **22** is provided with a screw hole **221** for fixing the one end **311** of the leaf spring **31** with a bolt or the like. The leaf spring **31** is an example of an elastic member included in the reaction force generation mechanism **30**. That is, the second housing **20** is configured to support a portion of the elastic member of the reaction force generation mechanism **30** opposite to the pedal pad **40**.

(124) Further, the second housing **20** extends from the spring receiving portion **22** toward the front of the vehicle, and has the portion **23** fixed to the vehicle body in a middle of such extension. The portion **23** fixed to the vehicle body extends respectively to a left-side and a right side, and is provided with a hole **24** piercing in a plate thickness direction. The second housing **20** is fixed to the floor **2** or the dash panel of the vehicle by inserting a bolt (not shown) through the hole **24**. The second housing **20** has one body construction in which the bearing portion **26**, the spring receiving portion **22** that supports the reaction force generation mechanism **30**, and the portion **23** fixed to the vehicle body are integrally formed by using the rigid portion **21**.

(125) The eighth embodiment described above also achieves the same effects as the sixth embodiment, the seventh embodiment or the like from the corresponding configuration as the sixth embodiment, the seventh embodiment or the like.

Ninth Embodiment

(126) Similarly to the above-described eighth embodiment, the ninth embodiment is a modification of the sixth or seventh embodiment. FIG. **21** shows only a second housing **20** included in a pedal device **1** of the ninth embodiment. The second housing **20** has a bearing portion **26** that supports a rotation shaft **41** of a pedal pad **40**.

(127) Further, the second housing **20** has a spring receiving portion **22** for supporting the one end **311** of a leaf spring **31** described in the first embodiment, at a portion on a vehicle body side with respect to the bearing portion **26**. The second housing **20** is configured to support a portion of the elastic member of the reaction force generation mechanism **30** opposite to the pedal pad **40**.

(128) The ninth embodiment described above can also achieve the same effects as the sixth to eighth embodiments and the like from the same corresponding configuration as the sixth to eighth embodiments and the like.

Tenth Embodiment

(129) As shown in FIG. **22**, in the tenth embodiment, a bearing member **46** that supports a rotation shaft **41** of a pedal pad **40** is configured to be directly attached to a floor **2** or a dash panel (that is, the vehicle body). A second housing **20** is configured to be attached to the floor **2** or the dash panel (that is, the vehicle body) at a position away from the bearing member **46**. The second housing **20** has a spring receiving portion **27** for installing a coil spring **32**. The coil spring **32** is an example of an elastic member of a reaction force generation mechanism **30**. That is, the second housing **20** supports a portion of the elastic member of the reaction force generation mechanism **30** on one side opposite to the pedal pad **40**.

(130) A first housing **10** covers the coil spring **32** as an elastic member of the reaction force generation mechanism **30** and the upper holder **37** provided at an upper end of the coil spring **32**. The first housing **10** has a largest opening **15** on one side facing the vehicle body, on which the pedal device **1** is installed. Further, the largest opening **15** of the first housing **10** is configured to be closed by the second housing **20**.

(131) The tenth embodiment described above also achieves the same effects as the first embodiment and the like from the corresponding configuration as the first embodiment and the like. Further, the pedal device **1** of the tenth embodiment is configured to simplify the

configurations of the first housing **10** and the second housing **20**.

Eleventh to Twelfth Embodiments

(132) In the eleventh to twelfth embodiments described below, part of the configuration of the second housing **20** is changed with respect to the fourth embodiment or the like, and other configurations are the same as those of the fourth embodiment, or the like. Therefore, only the part different from the fourth embodiment or the like is described.

Eleventh Embodiment

(133) As shown in FIG. **23**, a second housing **20** included in a pedal device **1** of the eleventh embodiment has a rigid portion **21** having a Young's modulus greater than that of a first housing **10**, and a resin portion **28** for molding the rigid portion **21**. The rigid portion **21** and the resin portion **28** of the second housing **20** are integrally formed by insert molding.

(134) The rigid portion **21** of the second housing **20** is arranged at a position between a portion of a reaction force generation mechanism **30** opposite to a pedal pad **40**, and a vehicle body. In the eleventh embodiment, the second housing **20** has (i) the rigid portion **21** arranged on one side facing the reaction force generation mechanism **30** and (ii) the resin portion **28** arranged on the other side facing the vehicle body. A spring receiving portion **27** for installing a coil spring **32** is provided on the rigid portion **21**. The coil spring **32** is an example of an elastic member of a reaction force generation mechanism **30**. That is, the rigid portion **21** of the second housing **20** supports a portion of the elastic member of the reaction force generation mechanism **30** opposite to the pedal pad **40**.

(135) The eleventh embodiment described above also achieves the same effects as the first embodiment or the like from the corresponding configuration as the first embodiment or the like. Further, the second housing **20** included in the pedal device **1** of the eleventh embodiment has the rigid portion **21** at least in part, and a portion of the reaction force generation mechanism **30** opposite to the pedal pad **40** is configured to be supported by the rigid portion **21**. Therefore, even when a driver applies a pedaling force to the pedal pad **40**, and a high load is input to the rigid portion **21** from the reaction force generation mechanism **30**, the amount of deformation of the rigid portion **21** is small. Thus, the reaction force generation mechanism **30** can generate a stable reaction force to the pedal pad **40**. In such manner, the pedal device **1** stabilizes the pedaling force characteristics and reduces the weight of the pedal device **1**, by adding the required-minimum members such as the rigid portion **21** to the portion **22** that supports the reaction force generation mechanism **30**. That is, the increase of the weight of the pedal device **1** is minimized for realizing the weight reduction.

Twelfth Embodiment

(136) As shown in FIG. **24**, a second housing **20** included in a pedal device **1** of the twelfth embodiment has a rigid portion **21** and a resin portion **28**. The rigid portion **21** and the resin portion **28** of the second housing **20** are integrally formed by insert molding.

(137) The rigid portion **21** of the second housing **20** is arranged at a position between a portion of a reaction force generation mechanism **30** opposite to a pedal pad **40**, and a vehicle body. In the twelfth embodiment, the second housing **20** has (i) the resin portion **28** arranged on one side facing the reaction force generation mechanism **30** and (ii) the rigid portion **21** arranged on the other side facing the vehicle body. A spring receiving portion **27** for installing a coil spring **32** is provided on the resin portion **28**. The coil spring **32** is an example of an elastic member that is provided in the reaction force generation mechanism **30**. The rigid portion **21** of the second housing **20** is arranged on a vehicle body side of the spring receiving portion **27** provided in the resin portion **28**. With such configuration, the rigid portion **21** of the second housing **20** supports a portion of the elastic member of the reaction force generation mechanism **30** opposite to the pedal pad **40**.

(138) As shown in FIG. **25**, the rigid portion **21** of the second housing **20** integrally has a portion **23** fixed to the vehicle body. In the second housing **20**, a portion (that is, the spring receiving portion **27**) that supports the reaction force generation mechanism **30** and the portion **23** that is

fixed to the vehicle body are integrally formed. The portion **23** fixed to the vehicle body extends from the rigid portion **21** to the left and right sides of the first housing **10**, and is provided with the hole **24** piercing in a plate thickness direction. The second housing **20** is fixed to a floor **2** or a dash panel of the vehicle by inserting a bolt (not shown) through the hole **24**.

(139) The twelfth embodiment described above also achieves the same effects as the first embodiment and the like from the corresponding configuration as the first embodiment and the like. Further, the second housing **20** included in the pedal device **1** of the twelfth embodiment also has the configuration, in which the rigid portion **21** is provided at least in part, and the portion of the reaction force generation mechanism **30** opposite to the pedal pad **40** is supported by the rigid portion **21**. Therefore, the pedal device **1** of the twelfth embodiment also achieves the same effects as those of the eleventh embodiment.

Thirteenth to Twenty-Second Embodiments

(140) The thirteenth to twenty-second embodiments described in the following respectively show a method of fixing a first housing **10** and a second housing **20** in contrast to the method in the first embodiment and the like, and the rest of the configuration is the same as the first embodiment and the like. Specifically, the thirteenth to fifteenth embodiments are examples of fixing methods using bolts **60** as fixing members. The sixteenth to eighteenth embodiments are examples of fixing methods by snap-fitting. The nineteenth embodiment is an example of a fixing method by caulking. The twentieth to twenty-second embodiments are examples in which a positioning structure is added to those fixing methods.

Thirteenth Embodiment

(141) As shown in FIG. **26**, a first housing **10** included in a pedal device **1** of the thirteenth embodiment has flange portions **18** respectively extending outward from portions of a left-side wall **12** and a right-side wall **13** of the first housing **10** on a side of a second housing **20**. The flange portion **18** is provided with a hole **19** piercing in a plate thickness direction. On the other hand, the second housing **20** is provided with a screw hole **29** at positions corresponding to the holes **19** of the flange portions **18**. A female screw (not shown) is provided on an inner wall of the screw hole **29**. The bolt **60** as a fixing member is inserted through the hole **19** of the flange portion **18** from the outside of the first housing **10** and screwed into the screw hole **29** of the second housing **20**. With such configuration, the first housing **10** and the second housing **20** are fixed.

(142) In the thirteenth embodiment described above, the bolt **60** as a fixing member is arranged at a position visible from an occupant when the pedal device **1** is in an attached state to the vehicle body. Therefore, the bolt **60** is removable while the pedal device **1** is attached to the vehicle body, and the first housing **10** and the second housing **20** are disassembled as required.

Fourteenth Embodiment

(143) As shown in FIG. **27**, a first housing **10** included in a pedal device **1** of the fourteenth embodiment has a threaded hole **101** opening toward a second housing **20** in a left-side wall **12** and a right-side wall **13** of the first housing **10**. A female screw (not shown) is provided on an inner wall of the threaded hole **101**. On the other hand, the second housing **20** is provided with a hole portion **201** piercing in a plate thickness direction at a position corresponding to the threaded hole **101** of the first housing **10**. The hole portion **201** provided in the second housing **20** is made of a small-diameter hole **202** on a first housing **10** side and a large-diameter hole **203** on a vehicle body side. The small-diameter hole **202** is sized to allow a shaft of a bolt **60** as a fixing member to pass through, and the large-diameter hole **203** is sized to accommodate a head of the bolt **60**. A step surface **204** is provided at a position between the small-diameter hole **202** and the large-diameter hole **203**.

(144) The bolt **60** as a fixing member is screwed into the threaded hole **101** of the first housing **10**, after passing through the large-diameter hole **203** and the small-diameter hole **202** from the outside of the second housing **20**. The head of the bolt **60** abuts on the step surface **204** while being accommodated in the large-diameter hole **203**. With such configuration, the first housing **10** and the

second housing **20** are fixed.

(145) In the fourteenth embodiment described above, the bolt **60** as a fixing member is arranged at a position invisible to the occupant when the pedal device **1** is attached to the vehicle body. Therefore, the bolt **60** cannot be removed while the pedal device **1** is attached to the vehicle body. As a result, the pedal device **1** is prevented from being unnecessarily disassembled.

Fifteenth Embodiment

(146) As shown in FIG. **28**, a second housing **20** provided in a pedal device **1** of the fifteenth embodiment has an upright wall portion **205** that extends upright from an extended portion extending outward from a first housing **10** along a left-side wall **12** and a right-side wall **13** of the first housing **10**. A hole **206** piercing through the upright wall portion **205** to reach the first housing **10** is provided in the upright wall portion **205**. On the other hand, the left-side wall **12** and the right-side wall **13** of the first housing **10** are respectively provided with a screw hole **102** at a position corresponding to the hole **206** of the upright wall portion **205**. A female thread (not shown) is provided on the inner wall of the screw hole **102**. The bolt **60** as a fixing member is inserted through the hole **206** of the upright wall portion **205** from the outside of the upright wall portion **205**, and is screwed into the screw hole **102** of the first housing **10**. With such configuration, the first housing **10** and the second housing **20** are fixed.

(147) In the fifteenth embodiment described above, in a state in which a pedal device **1** attached to a vehicle body, the bolt **60** as a fixing member is arranged at a position visible to the occupant. Therefore, the bolt **60** is removable while the pedal device **1** is attached to the vehicle body, and the first housing **10** and the second housing **20** can be disassembled as required.

Sixteenth Embodiment

(148) As shown in FIG. **29**, a first housing **10** provided in a pedal device **1** of the sixteenth embodiment has a locking claw **103** for snap fitting, which protrudes, toward a second housing **20**, from a surface of a left-side wall **12** and a right-side wall **13** facing the second housing **20**. A tip protrusion **104** is provided at a tip of the locking claw **103** and protrudes in a direction intersecting a facing direction in which the first housing **10** and the second housing **20** face each other.

(149) On the other hand, the second housing **20** is provided with the hole portion **201** piercing in a plate thickness direction at a position corresponding to the locking claw **103** of the first housing **10**. The hole portion **201** provided in the second housing **20** is made of a small-diameter hole **202** on a first housing **10** side and a large-diameter hole **203** on a vehicle body side. The small-diameter hole **202** is sized to allow the locking claw **103** to pass through, and the large-diameter hole **203** is sized to accommodate the tip protrusion **104** of the locking claw **103**. A step surface **204** is provided at a position between the small-diameter hole **202** and the large-diameter hole **203**.

(150) Further, the locking claw **103** protruding from the first housing **10** passes through the large-diameter hole **203** from the small-diameter hole **202** of the second housing **20**, and the tip protrusion **104** of the locking claw **103** engages with the step surface **204** in an accommodated state in the large-diameter hole **203**. With such configuration, the first housing **10** and the second housing **20** are fixed.

(151) In the sixteenth embodiment described above, by adopting a snap fitting to fix the first housing **10** and the second housing **20**, assemblability is improvable. Further, in the sixteenth embodiment, the snap-fitting locking claw **103** is arranged at a position invisible to the occupant when the pedal device **1** is attached to the vehicle body. Therefore, the snap fitting cannot be released while the pedal device **1** is attached to the vehicle body. Thus, the pedal device **1** is prevented from being unnecessarily disassembled.

Seventeenth Embodiment

(152) The seventeenth embodiment is a modification of the sixteenth embodiment. As shown in FIG. **30**, a first housing **10** included in a pedal device **1** of the seventeenth embodiment has a locking claw **103** for snap fitting. In the seventeenth embodiment, a tip protrusion **104** is provided substantially on the entire circumference of the locking claw **103**.

(153) In the seventeenth embodiment, the locking claw **103** protruding from the first housing **10** passes through a large-diameter hole **203** from a small-diameter hole **202** of a second housing **20**, and the tip protrusion **104** of the locking claw **103** engages with a step surface **204** in a state in which the tip protrusion **104** is housed in the large-diameter hole **203**. With such configuration, the first housing **10** and the second housing **20** are fixed.

(154) The seventeenth embodiment described above can thus have the effects similar to those of the sixteenth embodiment.

Eighteenth Embodiment

(155) As shown in FIG. **31**, a first housing **10** provided in a pedal device **1** of the eighteenth embodiment has a locking claw **103** for snap fitting, which protrudes outward in a middle of a left-side wall **12** and a right-side wall **13**. The locking claw **103** has an inclined shape in which the thickness gradually increases as the claw **103** recedes away from a second housing **20**. On the other hand, the second housing **20** has an upright wall portion **205** that rises along a left-side wall **12** and a right-side wall **13** of the first housing **10** from a portion that extends outward than the first housing **10**. A hole **210** for locking the locking claw **103** is provided at a portion of the upright wall portion **205** corresponding to the locking claw **103** of the first housing **10**.

(156) The locking claw **103** protruding from the left-side wall **12** and the right-side wall **13** of the first housing **10** engages with the hole **210** provided in the upright wall portion **205** of the second housing **20**, and is locked to a portion of an inner wall surface **211** of the hole **210** on one side opposite to the vehicle body. With such configuration, the first housing **10** and the second housing **20** are fixed.

(157) In the eighteenth embodiment described above, by adopting a snap fitting to fix the first housing **10** and the second housing **20** together, assemblability is improvable. Further, in the eighteenth embodiment, when the pedal device **1** is attached to the vehicle body, the locking claw **103** for snap-fitting is configured to be arranged at a position visible to the occupant. Therefore, the snap-fitted locking claw **103** is releasable in a state that the pedal device **1** is attached to the vehicle body, thereby allowing in such state to disassemble the first housing **10** and the second housing **20** as required.

Nineteenth Embodiment

(158) In the nineteenth embodiment, a first housing **10** and a second housing **20** are fixed by caulking. As shown in FIG. **32**, the first housing **10** provided in a pedal device **1** of the nineteenth embodiment has a column portion **105** and a large-diameter portion **106** protruding toward the second housing **20** from a surface of a left-side wall **12** and a right-side wall **13** facing a second housing **20**. The first housing **10**, the column portion **105**, and the large-diameter portion **106** are integrally formed of a continuous material. Also, the large-diameter portion **106** is formed to be greater than the column portion **105**. As indicated by a broken line **107** in FIG. **32**, the column portion **105** and the large-diameter portion **106** have the same size before the first housing **10** and the second housing **20** are fixed. When fixing the first housing **10** and the second housing **20** together, the large-diameter portion **106** is processed by thermal caulking to be greater than the column portion **105**.

(159) On the other hand, the second housing **20** is provided with a hole portion **201** piercing in a plate thickness direction of the second housing **20** at a position corresponding to the column portion **105** and the large-diameter portion **106** of the first housing **10**. The hole portion **201** provided in the second housing **20** is made of a small-diameter hole **202** on a first housing **10** side and a large-diameter hole **203** on a vehicle body side. The small-diameter hole **202** is sized to allow the column portion **105** to pass through, and the large-diameter hole **203** is sized to accommodate the large-diameter portion **106**. A step surface **204** is provided at a position between the small-diameter hole **202** and the large-diameter hole **203**.

(160) When assembling the first housing **10** and the second housing **20**, a protrusion indicated by the dashed line **107** in FIG. **32** is inserted through the hole portion **201** provided in the second

housing **20**. Then, the large-diameter portion **106** is formed by performing thermal caulking such that the tip of the projection is heated and crushed. Thereby, the first housing **10** and the second housing **20** are caulked and fixed.

(161) In the nineteenth embodiment described above, the first housing **10** and the second housing **20** are fixed by thermal caulking. Therefore, the first housing **10** and the second housing **20** are not disassembled unless the column portion **105** and the large-diameter portion **106** are cut. Further, in the nineteenth embodiment, when the pedal device **1** is attached to a vehicle body, the column portion **105** and the large-diameter portion **106** protruding from the first housing **10** are arranged at positions invisible to the occupant. Therefore, the pedal device **1** is prevented from being unnecessarily disassembled.

(162) Note that, as a modification of the nineteenth embodiment, although not shown in the drawing, the second housing **20** may be provided with a column portion and a large-diameter portion, and the second housing **20** and the first housing **10** may be fixed by caulking, by providing the first housing **10** with a hole portion.

Twentieth Embodiment

(163) The twentieth embodiment is an example in which a positioning structure is added to the fixing method described in the thirteenth embodiment. As shown in FIG. **33**, a positioning structure **62** provided in a pedal device **1** of the twentieth embodiment includes a first positioning portion **63** provided on a flange portion **18** of a first housing **10** and a second positioning portion **64** provided on a second housing **20**. In the twentieth embodiment, the first positioning portion **63** is a convex portion protruding from the flange portion **18** of the first housing **10** toward the second housing **20**. On the other hand, the second positioning portion **64** is a concave portion provided in the second housing **20** at a position corresponding to the first positioning portion **63**. By the engagement of the convex portion as the first positioning portion **63** and the concave portion as the second positioning portion **64**, an assembly position of the first housing **10** and the second housing **20** is determined with high accuracy. The positioning structure **62** may be provided at the same position where the bolt **60** serving as the fixing member is attached, or may be provided at a position different from where the bolt **60** is attached.

(164) FIGS. **34** to **36** show examples of the shapes of the first positioning portion **63** and the second positioning portion **64** that constitute the positioning structure **62**. As shown in FIG. **34**, the first positioning portion **63** may be columnar, and the second positioning portion **64** may correspondingly be cylindrical. As shown in FIGS. **35** and **36**, the first positioning portion **63** may be shaped like a square pole, and the second positioning portion **64** may be shaped like a corresponding rectangular tube. Although illustration is omitted, various configurations such as pins or ribs can be adopted as the convex portion forming the positioning structure **62**, and various configurations such as grooves or holes can be adopted as the concave portion forming the positioning structure **62**.

(165) In the twentieth embodiment described above, the assembly positions of the first housing **10** and the second housing **20** are accurately determined by the positioning structure **62**. According to the above, the pedal device **1** is provided with the positioning structure **62**, thereby having an improved accuracy for the assembly positioning for assembling the first housing **10** and the second housing **20**, compared with the assembling of the first housing **10** and the second housing **20** only with the bolt **60**.

(166) In the twentieth embodiment, the first housing **10** and the second housing **20** are fixed by using the bolts **60**. However, the fixing method is not limited to the above, and other methods such as snap-fitting, caulking, or the like may also be used.

Twenty-First Embodiment

(167) As shown in FIG. **37**, the twenty-first embodiment is an example in which a positioning structure **62** is added to the fixing method described in the fourteenth embodiment. Also, the positioning structure **62** described in the twenty-first embodiment is a modification of the

positioning structure **62** described in the twentieth embodiment. The positioning structure **62** provided in a pedal device **1** of the twenty-first embodiment includes a first positioning portion **63** provided on a first housing **10** and a second positioning portion **64** provided on a second housing **20**. In the twenty-first embodiment, the first positioning portion **63** is a concave portion provided in the first housing **10**. On the other hand, the second positioning portion **64** is a convex portion provided on the second housing **20** at a position corresponding to the first positioning portion **63**. By the engagement of the concave portion as the first positioning portion **63** and the convex portion as the second positioning portion **64**, the assembly position of the first housing **10** and the second housing **20** is determined with high accuracy. The positioning structure **62** may be provided at the same position where the bolt **60** serving as the fixing member is attached, or may be provided at a position different from the position where the bolt **60** is attached.

(168) In the twenty-first embodiment described above, the assembly position of the first housing **10** and the second housing **20** is accurately determined by the positioning structure **62**. According to the above, the pedal device **1** is provided with the positioning structure **62**, thereby having an improved accuracy for the assembly positioning for assembling the first housing **10** and the second housing **20**, compared with the assembling of the first housing **10** and the second housing **20** only with the bolt **60**.

(169) In the twenty-first embodiment, the bolt **60** is used as a fixing member in the method of fixing the first housing **10** and the second housing **20**. However, the method is not limited to the above, and snap-fitting, caulking, or the like may also be used.

Twenty-Second Embodiment

(170) An another example of a positioning structure **62** is described in the twenty-second embodiment. As shown in FIGS. **38** and **39**, in the twenty-second embodiment, a positioning structure **62** is configured as (i) a first positioning portion **65** provided on a left-side wall **12** or a right-side wall **13** of a first housing **10** and (ii) a second positioning portion **66** provided on an upright wall portion **205** of a second housing **20**. In the twenty-second embodiment, the first positioning portion **65** is a convex portion that protrudes outward from the left-side wall **12** or the right-side wall **13** of the first housing **10**. On the other hand, the second positioning portion **66** is a concave portion provided on the upright wall portion **205** of the second housing **20**. By the engagement of the convex portion as the first positioning portion **65** and the concave portion as the second positioning portion **66**, the assembly position of the first housing **10** and the second housing **20** is accurately determinable.

(171) The second embodiment described above can thus have the effects similar to those of the first embodiment.

Twenty-Third Embodiment

(172) As shown in FIG. **40**, a pedal device **1** of the twenty-third embodiment is configured by integrating a brake pedal device **3** and an accelerator pedal device **4**. The brake pedal device **3** and the accelerator pedal device **4** are installed in a common second housing **20**. The accelerator pedal device **4** is of an accelerator-by-wire type. The accelerator-by-wire system is a system that controls the opening and closing of an electronically-controlled throttle or the driving of a driving motor by an ECU mounted on a vehicle based on an electric signal output from a sensor unit that detects the swing angle of an accelerator pedal **400**. Note that the swing angle of the accelerator pedal **400** may be rephrased as a stroke amount of the accelerator pedal **400**.

(173) The configuration of the brake pedal device **3** is substantially the same as that described in the first embodiment. However, a second housing **20** of the brake pedal device **3** has a fixing portion **200** that extends outside an outer edge of a first housing **10** of the brake pedal device **3** and supports a housing **401** of the accelerator pedal device **4**. A portion **230** of the second housing **20** that supports the brake pedal device **3** and the fixed portion **200** that supports the accelerator pedal device **4** are integrally formed of a continuous material.

(174) Also in the twenty-third embodiment, the second housing **20** is made of a member having a

Young's modulus greater than that of the first housing **10** of the brake pedal device **3** as in the first embodiment, or has a member having a Young's modulus greater than that of the first housing **10** at least in part. The second housing **20** supports a portion of the reaction force generation mechanism **30** of the brake pedal device **3** on one side opposite to the pedal pad **40** with the rigid portion **21**.

(175) In the twenty-third embodiment described above, the second housing **20** has the fixing portion **200** that extends outside the outer edge of the first housing **10** of the brake pedal device **3** and supports the accelerator pedal device **4**. According to the above, it is possible to integrate the brake pedal device **3** and the accelerator pedal device **4** into one body. Therefore, the pedal pad **40** of the brake pedal device **3** and the accelerator pedal **400** of the accelerator pedal device **4** can be assembled to the vehicle while maintaining the positional relationship between them. Specifically, the brake pedal device **3** is assembled to the vehicle in a state, in which a positional relationship is maintained to have the accelerator pedal **400** positioned farther away from the pedal pad **40** of the brake pedal device **3**. Therefore, the assemblability of the brake pedal device **3** and the accelerator pedal device **4** is improved while the safety of the vehicle is enhanced. Further, the brake pedal device **3** and the accelerator pedal device **4** is easily attachable to the vehicle as one component.

Other Embodiments

(176) (1) In each of the above-described embodiments, the brake pedal device is described as an example of the pedal device **1**, but the pedal device **1** is not limited to the above, and can also be an accelerator pedal device. Alternatively, the pedal device **1** can be various devices that are operated by the driver's foot.

(177) (2) In each of the above-described embodiments, as an example of the pedal device **1**, the pedal pad **40** and the master cylinder **126** are not mechanically connected, but the present disclosure is not limited to the above. For example, the pedal device **1** may be a device in which the pedal pad **40** and the master cylinder **126** are mechanically connected.

(178) (3) In each of the above-described embodiments, an example of the reaction force generation mechanism **30** is described as a combination of the leaf spring **31** and a plurality of the coil springs **33** and **34**, or as a single coil spring **32**. However, the reaction force generation mechanism **30** is not limited to the above. For example, the reaction force generation mechanism **30** may also be made of a plurality of the coil springs **33** and **34**, or may also be made of one or a plurality of the leaf springs **31**. Alternatively, the pedal pad **40** and a master cylinder **126** may be mechanically connected and a reaction force against the pedaling force applied to the pedal pad **40** by the driver may be generated by the master cylinder **126**.

(179) (4) In each of the above-described embodiments, the brake-by-wire system **100** has a configuration in which the master cylinder **126** generates hydraulic pressure in the brake fluid flowing through the brake circuit **120**, but the present disclosure is not limited to the above. For example, the brake-by-wire system **100** may be configured to generate hydraulic pressure in the brake fluid flowing through the brake circuit **120** using a hydraulic pump.

(180) (5) In the first embodiment, the ECU **110** is configured by the first ECU **111** and the second ECU **112**, but the present disclosure is not limited to the above. For example, one or more than three ECUs **110** may be used.

(181) (6) In the eleventh and twelfth embodiments, the configuration in which the second housing **20** has the rigid portion **21** as part of the resin portion **28** has been described, but the configuration is not limited to the above. For example, the rigid portion **21** of the second housing **20** may be embedded in the resin portion **28**.

(182) The present disclosure is not limited to the above-described embodiments, and can be appropriately modified. The above-described embodiments are not independent of each other, and can be appropriately combined together except when the combination is obviously impossible. Further, individual elements or features of a particular embodiment are not necessarily essential unless it is specifically stated that the elements or the features are essential in the foregoing description, or unless the elements or the features are obviously essential in principle. Further, in

each of the embodiments described above, when numerical values such as the number, numerical value, quantity, range, and the like of the constituent elements of the embodiment are referred to, except for the cases where the numerical values are expressly indispensable in particular, and the cases where the numerical values are obviously limited to a specific number in principle, and the like, the present disclosure is not limited to such value or number. Further, a shape, positional relationship or the like of a configuration element, which is referred to in the embodiments described above, is not necessarily limiting the disclosure, unless it is specifically described so or obviously necessary in principle.

Claims

1. A pedal device configured to transmit an electric signal corresponding to an amount of a pedal operation by a driver to an electronic control device of a vehicle, the pedal device comprising: a pedal pad configured to swing about a predetermined axis when being stepped on by a driver's foot, wherein a step portion of the pedal pad to be stepped on by the driver is arranged above the axis in a vertical direction when mounted on the vehicle; a sensor unit configured to output a signal corresponding to a swing angle of the pedal pad; a reaction force generation mechanism having an elastic member configured to generate a reaction force against a driver's stepping force applied to the pedal pad; a first housing configured to hold or cover at least one of a rotation shaft provided at the axis of the swinging of the pedal pad, the sensor unit, and the reaction force generation mechanism; and a second housing made of a portion having a Young's modulus greater than that of the first housing or configured to have the portion at least in part, wherein the second housing is configured to support a portion of the reaction force generation mechanism on one side opposite to the pedal pad by the portion, wherein the first housing or the second housing has a bearing portion rotatably supporting the rotation shaft of the pedal pad.
2. The pedal device according to claim 1, wherein the portion of the second housing is disposed at a position between the portion of the reaction force generation mechanism opposite to the pedal pad, and a vehicle body.
3. The pedal device according to claim 1, wherein the second housing is formed in one body integrally by the portion of the second housing, to include: a first portion that supports the reaction force generation mechanism; and a second portion that is fixed to the vehicle body.
4. The pedal device according to claim 1, wherein the portion of the second housing is made of metal.
5. The pedal device according to claim 1, wherein the first housing has a largest opening into which an elastic member of the reaction force generation mechanism is insertable without bending on one side where the second housing is provided, and the largest opening is closed by the second housing.
6. The pedal device according to claim 1, wherein the elastic member includes at least one leaf spring, and the leaf spring is fixed to the portion of the second housing.
7. The pedal device according to claim 1, wherein the first housing and the second housing are fixed by a fixing member, and the fixing member is fixed to the first housing through a hole provided in the second housing from a position on a vehicle body side in a state in which the pedal device is attached to the vehicle body.
8. The pedal device according to claim 1, wherein the first housing and the second housing are fixed by snap fitting using a locking claw, and the locking claw of the snap fitting is arranged at a position invisible to an occupant when the pedal device is attached to the vehicle body.
9. The pedal device according to claim 1, wherein a hole portion is provided in one of the first housing or the second housing, a column portion is provided to protrude from an another one of the first housing or the second housing and to pass through the hole portion, and a large-diameter portion is integrally formed of a continuous material with the column portion and made to be greater than the hole portion, and the first housing and the second housing are caulked and fixed by

using the hole portion, the column portion and the larger-diameter portion.

10. The pedal device according to claim 1, wherein the first housing is provided with a first positioning portion, and the second housing is provided with a second positioning portion, and the first positioning portion and the second positioning portion are fitted to set an assembly position of the first housing and the second housing.

11. The pedal device according to claim 1, wherein the pedal device is one of an accelerator pedal device or a brake pedal device of the vehicle, and the second housing has a fixing portion that extends outside than an outer edge of the first housing to support an other one of the accelerator pedal device or the brake pedal device.

12. The pedal device according to claim 1, wherein the pedal device is a brake pedal device used in a brake-by-wire system in which a brake circuit generates a hydraulic pressure required for braking the vehicle by a drive control of the electronic control device based on an output signal of the sensor unit.

13. A pedal device configured to transmit an electric signal corresponding to an amount of a pedal operation by a driver to an electronic control device of a vehicle, the pedal device comprising: a pedal pad configured to swing about a predetermined axis when being stepped on by a driver's foot, wherein a step portion of the pedal pad to be stepped on by the driver is arranged above the axis in a vertical direction when mounted on the vehicle; a sensor unit configured to output a signal corresponding to a swing angle of the pedal pad, a reaction force generation mechanism having an elastic member configured to generate a reaction force against a driver's stepping force applied to the pedal pad, a first housing configured to hold or cover at least one of a rotation shaft provided at the axis of the swinging of the pedal pad, the sensor unit, and the reaction force generation mechanism; and a second housing made of a portion having a Young's modulus greater than that of the first housing or configured to have the portion at least in part, wherein the second housing is configured to support a portion of the reaction force generation mechanism on one side opposite to the pedal pad by the portion, wherein the first housing has an opening into which the reaction force generation mechanism is insertable in at least one direction, and covers the reaction force generation mechanism together with the second housing that closes the opening.

14. A pedal device configured to transmit an electric signal corresponding to an amount of a pedal operation by a driver to an electronic control device of a vehicle, the pedal device comprising: a pedal pad configured to swing about a predetermined axis when being stepped on by a driver's foot, wherein a step portion of the pedal pad to be stepped on by the driver is arranged above the axis in a vertical direction when mounted on the vehicle; a sensor unit configured to output a signal corresponding to a swing angle of the pedal pad, a reaction force generation mechanism having an elastic member configured to generate a reaction force against a driver's stepping force applied to the pedal pad; a first housing configured to hold or cover at least one of a rotation shaft provided at the axis of the swinging of the pedal pad, the sensor unit, and the reaction force generation mechanism; and a second housing made of a portion having a Young's modulus greater than that of the first housing or configured to have the portion at least in part, wherein the second housing is configured to support a portion of the reaction force generation mechanism on one side opposite to the pedal pad by the portion, wherein the reaction force generation mechanism has a plurality of the elastic members.
